

Jieke Shi

Research Scientist | NRF Postdoctoral Award Recipient
Singapore Management University

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RESEARCH INTERESTS

Software Engineering, Program Synthesis, AI System Testing, Code LLMs

I design and implement **automated synthesis**-assisted methods for validating the **safety** of AI-enabled system and enhancing the **efficiency** of large language models serving.

EDUCATION

Aug 2022 – Mar 2025 **Singapore Management University**, School of Computing and Information Systems
Ph.D. Candidate in Computer Science

- **31 Conference/Journal Publications**
- **National Research Foundation (NRF) Postdoctoral Award** (\$250,000)
- **ACM SIGSOFT Distinguished Paper Awards** (ASE'22 nomination, ASE'25 winner, ICSE'26 winner)
- **IEEE Computer Society TCSE Distinguished Paper Award** (SANER 2026)
- **IEEE Computer Society Best Paper Award** (2025)
- **SMU Research Staff Excellence Awards** (2025)
- **SMU SCIS Dean's List Award** (2023 & 2025)
- **SMU Presidential Doctoral Fellowship** (2023)
- **Honorable Mention Award** (ACSAC 2022)

Supervisor: ACM/IEEE/ASE Fellow, OUB Chair Professor [David Lo](#)

Aug 2016 – Jun 2020 **Yangzhou University**, School of Information Engineering
B.Eng. in Electronic Engineering

- **Outstanding Undergraduate Thesis Award** (Top 0.2% in province)
- **Outstanding Graduate** (Top 5%)

EMPLOYMENT

Mar 2026 – Present **Singapore Management University**, Research Scientist

Awarded the National Research Foundation (NRF) Postdoctoral Award, which provides up to 4 years of funding with a SGD 250,000 research grant to support independent research.

Jun 2021 – Mar 2026 **Singapore Management University**, Research Engineer

Working as lead research staff under the AI.SG Programme on the testing of Reinforcement Learning (RL)-enabled systems. Led to multiple publications covering runtime shielding and RL system safety validation, including [J.5, J.7, C.5, C.8, C.9].

Jul 2020 – Apr 2021 **Chinese Academy of Sciences, Institute of Computing**, Research Assistant Intern

Developed a prototype system for generating faithful natural language text from structured data.

PUBLICATIONS

Note: In software engineering, ICSE and ASE are recognized as top-tier conferences, and TOSEM and TSE are top-tier journals (CORE A*); MSR, ICSME, and SANER are prominent conferences covering software mining, analysis, and evolution (CORE A). In security, S&P is a top-tier conference (CORE A*), while ACSAC is a leading venue with a focus on system security applications (CORE A).

[C]: Full Conference Paper (Total: 17) [J]: Full Journal Paper (Total: 8) [S]: Short Paper (Total: 6)

2026

- [C.17] Chengyan Ma, **Jieke Shi**, Ruidong Han, Ye Liu, Feng Li, Yuqing Niu, and David Lo. “Automated Repair of TEE Partitioning Issues via DSL-Guided and LLM-Assisted Patching”. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 27 pages.
- [C.16] **Jieke Shi**, Junda He, Zhou Yang, Chengran Yang, Mykhailo Klymenko, Thong Hoang, Sherry Xu, Zhenchang Xing, and David Lo. “Towards Efficient Code LLM Inference with Quantization and Compilation-Time Optimization”. In Proceedings of the 34th ACM International Conference on the Foundations of Software Engineering (**FSE 2026**, Research Track), 27 pages.
- [S.6] Chengyan Ma, **Jieke Shi**, Ruidong Han, Ye Liu, Yuqing Niu, and David Lo. “Finding Missing Input Validation in TEEs via LLM-Assisted Symbolic Execution”. In Proceedings of the 3rd ACM International Conference on AI Foundation Models and Software Engineering (**FORGE 2026**, New Idea Papers).
- [J.8] Junda He, **Jieke Shi**, Terry Yue Zhuo, Christoph Treude, Jiamou Sun, Zhenchang Xing, Xiaoning Du, and David Lo. “LLM-as-a-Judge for Software Engineering: Literature Review, Vision, and the Road Ahead”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2026**). [\[PDF\]](#)
- [C.15] Yuqing Niu, **Jieke Shi**, Ruidong Han, Ye Liu, Chengyan Ma, Yunbo Lyu, and David Lo. “What You Trust Is Insecure: Demystifying How Developers (Mis)Use Trusted Execution Environments in Practice”. In Proceedings of the 33rd IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2026**, **IEEE Computer Society TCSE Distinguished Paper Award**). [\[PDF\]](#) [\[Code\]](#)
- [S.5] Huihui Huang, **Jieke Shi**, Junkai Chen, Ting Zhang, Yikun Li, Chengran Yang, Eng Lieh Ouh, Lwin Khin Shar, and David Lo. “PenForge: On-the-Fly Expert Agent Construction for Automated Penetration Testing”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, NIER Track). [\[PDF\]](#) [\[Code\]](#)
- [C.14] Huihui Huang, Ratnadira Widyasari, Ting Zhang, Ivana Clairine IRSAN, **Jieke Shi**, Han Wei Ang, Frank Liauw, Eng Lieh Ouh, Lwin Khin Shar, Hong Jin Kang, and David Lo. “Back to the Basics: Rethinking Issue-Commit Linking with LLM-Assisted Retrieval”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**). [\[PDF\]](#) [\[Code\]](#)
- [C.13] Chengran Yang, Zhensu Sun, Hong Jin Kang, **Jieke Shi**, and David Lo. “Think Like Human Developers: Harnessing Community Knowledge for Structured Code Reasoning”. In Proceedings of the 48th IEEE/ACM International Conference on Software Engineering (**ICSE 2026**, **ACM SIGSOFT Distinguished Paper Award**). [\[PDF\]](#)

2025

- [J.7] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, Dongsun Kim, DongGyun Han, and David Lo. “Finding Safety Violations of AI-Enabled Control Systems through the Lens of Synthesized Proxy Programs”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#) [\[Code\]](#)
- [J.6] **Jieke Shi**, Zhou Yang, and David Lo. “Efficient and Green Large Language Models for Software Engineering: Literature Review, Vision, and the Road Ahead”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)
- [J.5] **Jieke Shi**, Junda He, Zhou Yang, Đorđe Žikelić, and David Lo. “Synthesizing Efficient and Permissive Programmatic Runtime Shields for Neural Policies”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)
- [J.4] Zhou Yang, **Jieke Shi**, Premkumar Devanbu, and David Lo. “Ecosystem of Large Language Models for Code”. ACM Transactions on Software Engineering and Methodology (**TOSEM 2025**). [\[PDF\]](#)

- [C.12] Yunbo Lyu, Zhou Yang, **Jieke Shi**, Jianming Chang, Yue Liu, and David Lo. “My productivity is boosted, but ...” *Demystifying Users’ Perception on AI Coding Assistants*. In Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2025**, 🏆 **ACM SIGSOFT Distinguished Paper Award**). [PDF] [Code]

2024

- [C.11] **Jieke Shi**, Zhou Yang, Hong Jin Kang, Bowen Xu, Junda He, and David Lo. “Greening Large Language Models of Code”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]
- [J.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, Bowen Xu, Xin Zhou, DongGyun Han, and David Lo. “Prioritizing Speech Test Cases”. *ACM Transactions on Software Engineering and Methodology* (**TOSEM 2024**). [PDF] [Code]
- [J.2] Zhou Yang, Bowen Xu, Jie M. Zhang, Hong Jin Kang, **Jieke Shi**, Junda He, and David Lo. “Stealthy Backdoor Attack for Code Models”. *IEEE Transactions on Software Engineering* (**TSE 2024**, 🏆 **IEEE Computer Society Best Paper Award**). [PDF] [Code]
- [J.1] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Gotcha! This Model Uses My Code! Evaluating Membership Leakage Risks in Code Models”. *IEEE Transactions on Software Engineering* (**TSE 2024**). [PDF]
- [C.10] Chen Gong, Zhou Yang, Yunpeng Bai, Junda He, **Jieke Shi**, Kecen Li, Arunesh Sinha, Bowen Xu, Xinwen Hou, David Lo, and Tianhao Wang. “BAFFLE: Hiding Backdoors in Offline Reinforcement Learning Datasets”. In Proceedings of the 45th IEEE Symposium on Security and Privacy (**S&P 2024**). [PDF] [Code]
- [C.9] Junda He, Zhou Yang, **Jieke Shi**, Chengran Yang, Kisub Kim, Bowen Xu, Xin Zhou, and David Lo. “Curiosity-Driven Testing for Sequential Decision-Making Process”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]
- [C.8] Zhou Yang, Zhipeng Zhao, Chenyu Wang, **Jieke Shi**, Dongsun Kim, DongGyun Han, and David Lo. “Unveiling Memorization in Code Models”. In Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (**ICSE 2024**). [PDF] [Code]

2023

- [C.7] Zhou Yang, Chenyu Wang, **Jieke Shi**, Thong Hoang, Pavneet Kochhar, Qinghua Lu, Zhenchang Xing, and David Lo. “What Do Users Ask in Open-Source AI Repositories? An Empirical Study of GitHub Issues”. In Proceedings of the 20th IEEE/ACM International Conference on Mining Software Repositories (**MSR 2023**). [PDF] [Code]

2022

- [C.6] **Jieke Shi**, Zhou Yang, Bowen Xu, Hong Jin Kang, and David Lo. “Compressing Pre-trained Models of Code into 3 MB”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**, 🏆 **ACM SIGSOFT Distinguished Paper Award Nomination**). [PDF] [Poster] [Code]
- [S.2] **Jieke Shi**, Zhou Yang, Junda He, Bowen Xu, and David Lo. “Can Identifier Splitting Improve Open-Vocabulary Language Model of Code?”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**, ERA Track). [PDF] [Poster] [Code]
- [C.5] Chen Gong, Zhou Yang, Yunpeng Bai, **Jieke Shi**, Arunesh Sinha, Bowen Xu, David Lo, Xinwen Hou, Guoliang Fan. “Curiosity-Driven and Victim-Aware Adversarial Policies”. In Proceedings of the 39th Annual Computer Security Applications Conference (**ACSAC 2022**, 🏆 **Honorable Mention Award**). [PDF] [Code]

- [C.4] Chengran Yang, Bowen Xu, Ferdian Thung, Yucen Shi, Ting Zhang, Zhou Yang, Xin Zhou, **Jieke Shi**, Junda He, DongGyun Han, and David Lo. “*Answer Summarization for Technical Queries: Benchmark and New Approach*”. In Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2022**). [[PDF](#)] [[Code](#)]
- [C.3] Zhou Yang, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “*Revisiting Neuron Coverage Metrics and Quality of Deep Neural Networks*”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [[PDF](#)] [[Code](#)]
- [C.2] Ratnadira Widyasari, Stefanus Agus Haryono, Ferdian Thung, **Jieke Shi**, Constance Tan, Fiona Wee, Jack Phan, and David Lo. “*On the Influence of Biases in Bug Localization: Evaluation and Benchmark*”. In Proceedings of the 29th IEEE International Conference on Software Analysis, Evolution and Reengineering (**SANER 2022**). [[PDF](#)] [[Code](#)] [[Dataset](#)]
- [C.1] Zhou Yang, **Jieke Shi**, Junda He, and David Lo. “*Natural Attack for Pre-trained Models of Code*”. In Proceedings of the 44th IEEE/ACM International Conference on Software Engineering (**ICSE 2022**). [[PDF](#)] [[Code](#)]

2021

- [S.1] Zhou Yang 🏆, **Jieke Shi** 🏆, Shaowei Wang, and David Lo. “*IncBL: Incremental Bug Localization*”. In Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering (**ASE 2021**, Tool Demonstrations). [[PDF](#)] [[Poster](#)] [[Code](#)]
- [S.2] Zhou Yang, Harshit Jain, **Jieke Shi**, Muhammad Hilmi Asyrofi, and David Lo. “*BiasHeal: On-the-Fly Black-Box Healing of Bias in Sentiment Analysis Systems*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2021**, NIER Track). [[PDF](#)] [[Code](#)]
- [S.3] Muhammad Hilmi Asyrofi, Zhou Yang, **Jieke Shi**, Chu Wei Quan, and David Lo. “*Can Differential Testing Improve Automatic Speech Recognition Systems?*”. In Proceedings of the 37th IEEE International Conference on Software Maintenance and Evolution (**ICSME 2021**, NIER Track). [[PDF](#)] [[Code](#)]

HONORS & AWARDS

- 2026 **National Research Foundation (NRF) Postdoctoral Award** (\$\$250,000), Singapore NRF
ACM SIGSOFT Distinguished Paper Award, ICSE 2026
IEEE Computer Society TCSE Distinguished Paper Awards, SANER 2026
- 2025 **ACM SIGSOFT Distinguished Paper Award**, ASE 2025
IEEE Computer Society Best Paper Award, IEEE Transactions on Software Engineering
OpenAI Researcher Access Program Grant (\$5,000)
Google Cloud Research Credits Grant (\$2,500)
Research Staff Excellence Awards (awarded to only 4 researchers university-wide), Singapore Management University
SMU SCIS PhD Research Excellence Award, Singapore Management University
SMU SCIS Dean’s List Award, Singapore Management University
- 2023 **Presidential Doctoral Fellowship**, Singapore Management University
SCIS Dean’s List Award, Singapore Management University
- 2022 **Honorable Mention Award**, ACSAC 2022
ACM SIGSOFT Distinguished Paper Award Nomination, ASE 2022
- 2020 **Outstanding Undergraduate Thesis** (Top 0.2% in province), Jiangsu Province, China
Outstanding Graduate (Top 5%), Yangzhou University
- 2019 & Before **National Encouragement Scholarship** (Top 3%), Ministry of Education of China

2nd Class Presidential Scholarship, Yangzhou University
1st Class Presidential Scholarship, Yangzhou University

ACADEMIC SERVICES

Program Committee (PC)/Reviewer

- 2026 **International Conference on the Foundations of Software Engineering (FSE)**, PC in Ideas, Visions and Reflections Track
- 2026 **Workshop on Human-Centered AI for Software Engineering (HumanAISE)**, PC
- 2026 **International Conference on AI-powered Software (Alware)**, PC in Main Track
- 2026 **International Conference on Software Maintenance and Evolution (ICSME)**, PC in Research Papers & Visions and Emerging Results Track
- 2025–2026 **International Conference on Software Engineering (ICSE)**, Shadow PC
- 2024–2026 **Mining Software Repositories Conference (MSR)**, Junior PC
- 2025–2026 **International Workshop on Green and Sustainable Software (GREENS@ICSE)**, PC
- 2026 **European Conference on Computer Systems (EuroSys)**, Shadow PC & Artifact Evaluation
- 2026 **International Conference on Evaluation and Assessment in Software Engineering (EASE)**, PC in AI Models/Data Track
- 2025 **Annual Meeting of the Association for Computational Linguistics (ACL)**, Reviewer
- 2025 **IEEE International Computer Software and Applications Conference (COMPSAC)**, PC

Organizing Committee

- 2024 **International Workshop on Software Engineering and AI for Data Quality in Cyber-Physical Systems (SEA4DQ@FSE)**, Web Chair

Journal Reviewer

- 2025–Present **IEEE Transactions on Dependable and Secure Computing (TDSC)**
- 2025–Present **Empirical Software Engineering (EMSE)**
- 2023–Present **ACM Transactions on Software Engineering and Methodology (TOSEM)**
- 2023–Present **IEEE Transactions on Software Engineering (TSE)**
- 2021–Present **Neurocomputing (Software Section)**

External Reviewer

- 2021–2025 **International Conference on Software Engineering (ICSE)**
- 2022–2026 **IEEE/ACM International Conference on Automated Software Engineering (ASE)**
- 2023–2025 **ACM International Conference on the Foundations of Software Engineering (ESEC/FSE)**
- 2023 **International Conference on Software Maintenance and Evolution (ICSME)**
- 2023 **International Workshop on LLMs for Code (LLM4Code@ICSE)**
- 2022–2023 **International Joint Conference on Artificial Intelligence (IJCAI)**
- 2022 **IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER)**
- 2022 **International Conference on AI Engineering – Software Engineering for AI (CAIN)**
- 2022 **International Workshop on Software Engineering for Responsible AI (SE4RAI@ICSE)**
- 2022 **International Conference on Engineering of Complex Computer Systems (ICECCS)**

INVITED TALKS

- Dec 2025 **Shanghai Jiao Tong University**, Research Seminar
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis
- Nov 2025 **A-Mobile Workshop of ASE 2025**, Keynote, Invited by Guozhu Meng
Optimizing and Fortifying AI Software through the Lens of Artifact Synthesis
- Nov 2025 **Korea University**, Workshop on the Next Generation of CodeLLMs and Their Applications, Invited by Dongsun Kim
Optimizing and Fortifying AI Software via Program Synthesis
- Sep 2025 **Singapore Management University**, RISE-WASP Workshop, Invited by Lingxiao Jiang
Fortifying Autonomous Systems via Program Synthesis
- Nov 2023 **Singapore Management University**, Research Cluster Seminar, Invited by David Lo
Towards Compact and Efficient Language Models of Code via Compression
- Oct 2023 **North Carolina State University**, Guest Lecture, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [[Link](#)]

TEACHING EXPERIENCE

- Jan 2026 **Singapore Management University**, Teaching Assistant in CS706, Invited by David Lo
Software Mining and Analysis, Graduate Core Course
- Oct 2023 **North Carolina State University**, Guest Lecture in CSC791, Invited by Bowen Xu
Enhancing Efficiency of Large Language Models of Code via Model Compression [[Link](#)]
- Jan 2016 - Jun 2016 **Yangzhou University**, Teaching Assistant
Digital Signal Processing, Undergraduate Core Course

GRANT PROPOSAL WRITING EXPERIENCE

- 2025 - 2027 **ESG-based Responsible AI: Toward Green, Secure, and Compliant LLM Utilisation for Digital Service Development Process**
CSIRO - A*STAR Partnership Program, Singapore & Australia
Total Award: Over SGD \$500K
- 2024 - 2028 **TrustedSEERs: Trusted Intelligent Work Bots for Engineering Better Software**
National Research Foundation (NRF), Singapore
Total Award: Over SGD \$3.2M
- 2023 - 2025 **TitanCA: Vulnerability Detection using Large Code Models**
Government Technology Agency (GovTech), Singapore
Total Award: Over SGD \$3.8M

REFEREES

David Lo, ACM/IEEE/ASE Fellow, ACM Distinguished Speaker, OUB Chair Professor
School of Computing and Information Systems, Singapore Management University
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Zhou Yang, Canada CIFAR AI Chair Nominee, Amii Fellow, Assistant Professor
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